

Board of Studies (BoS) Syllabus Template

Course Information

Degree: UG

Program: BCA

Course Name: Environmental Science and Sustainability

Course Credits: [2]

Semester: 1st Semester

Course Category: Core

Type of Course: Lecture

New Course: No

Learning Hours

Activity	Hours per Week
Lecture	2
Tutorial	0
Practical	0
Experiential Learning	0

Course Contribution

Category	Percentage Contribution
Employment Enhancement	30%
Entrepreneurship Skills	20%
Skill Development	50%

Course Relevance

Level	Percentage Relevance
Global	25%
National	40%
Regional	40%
Local	60%

Course Outcomes (COs)

CO#	Course Outcome	Bloom's Taxonomy Level
C01	Discuss with students the importance of Environment and to protect the eco systems mainly forests, rivers and lakes etc	BL-2
C02	Demonstrate on sustainable development by reducing the use of non-renewable resources and increasing renewable energy resources with the support of biodiversity services	BL-3
C03	Criticize on the harmful effects of various pollution to people around and help to reduce pollution	BL-4
C04	Discriminate the importance of global warming, ozone layer depletion and its impacts with the understanding of various environmental laws	BL-5
C05	Examine the resettlement and rehabilitation on environment, Support the government and the NGO's in protecting our valuable flora, fauna and endangered species.	BL-6

Module	Course Content	Tools Used / Assessment and Activity
Module 1: Fundamentals of Environment and Sustainability	• Introduction to environmental concepts and components and segments of the environment •	
	• Human-environment relationship and historical environmental movements	
	• Concept of sustainability and its significance	
	• Natural resource classification and overutilization issues	
	• Conservation strategies for natural resources (water, energy, forests, food security)	
	• Equity in resource usage: intergenerational and intragenerational perspectives	
Module 2: Ecosystems and Biodiversity Conservation	• Structure, function, and characteristics of ecosystems	
	• Ecological concepts: resilience, homeostasis, and carrying capacity	

	• Importance of biodiversity and associated threats. Strategies for biodiversity conservation: in situ & ex situ conservation	
	• Significance of nature reserves and biodiversity hotspots	
	• India as a mega-diverse nation and conservation policies	
Module 3: [Module Title]	• Types of environmental pollution: air, water, noise, soil, marine Causes and impacts of pollution on businesses and communities	
	• Climate change, ozone layer depletion, acid rain, and the greenhouse effect	
	• Sustainable waste management practices and cleaner technologies	
	• Natural and man-made disasters: prevention, management, and mitigation	
	• Business responsibility in disaster risk reduction	

Module 4: Environmental Legislation and Ethics	• Environmental justice and human rights (environmental refugees, rehabilitation)	
	• Role of businesses in sustainable development and ethical considerations • Key environmental laws: ◦ Water (Prevention and Control of Pollution) Act, 1974 ◦ Air (Prevention and Control of Pollution) Act, 1981 ◦ Environment (Protection) Act, 1986	
	• Relationship between society and environment	
	• Judiciary's role in environmental protection	
Module 5: Sustainable Development and Practical Applications	• Sustainable development goals (SDGs) and corporate responsibility	
	• Role of ecological economics in environmental decision-making	
	• Human population growth, demographic changes, and resource challenges. Case studies on sustainable business practices	
	• Public awareness and education for sustainability. Future trends in environmental management	

CO-PO/PSO Mapping

CO#	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	2	0	0	0	0	0	1	0	0
CO2	0	2	0	0	2	2	1	0	0
CO3	0	2	2	0	2	0	0	0	0
CO4	0	3	2	2	3	2	3	2	3
CO5	0	0	3	2	3	3	3	3	3